Solutions to

Subtitle or Series

Main Title of the Book

Author
Your Name

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Chapter 1.

Basic Problems

Problem 1. Roots of a quadratic

Solve the equation $x^2 - 5x + 6 = 0$.

Solution: We look for two numbers whose sum is 5 and whose product is 6: they are 2 and 3, so the polynomial factors as

$$x^2 - 5x + 6 = (x - 2)(x - 3). \tag{1.1}$$

A product is zero only when one of its factors is zero, so $x - 2 = 0$ or $x - 3 = 0$.

Answer to Problem 1.

$x = 2$ or $x = 3$.

Verification: Substitution gives $2^2 - 5 \cdot 2 + 6 = 0$ and $3^2 - 5 \cdot 3 + 6 = 0$. The graph in Fig. 1.1 crosses the x axis at these two roots.



Figure 1.1: The polynomial $y = x^2 - 5x + 6$ and its roots.

Problem 2.

Let $S_n = 1 + 3 + 5 + \cdots + (2n-1)$ be the sum of the first n odd numbers.

(a)

Compute S_1, S_2, S_3 and S_4, and guess a formula for S_n.

Solution: Adding one odd number at a time,

$$S_1 = 1, \qquad S_2 = 1 + 3 = 4, \qquad S_3 = 4 + 5 = 9, \qquad S_4 = 9 + 7 = 16.$$

These are the squares of 1, 2, 3 and 4.

Answer to Problem 2 (a).

$S_n = n^2$.

(b)

Prove the formula of part (a) by induction.

Solution: For $n = 1$, $S_1 = 1 = 1^2$. If $S_n = n^2$ for some $n \geq 1$, then

$$S_{n+1} = S_n + (2n+1) = n^2 + 2n + 1 = (n+1)^2, \tag{1.2}$$

so the formula also holds for $n+1$, and by induction for every $n \geq 1$. This part is a proof, so it has no answer box.

(c) Sum of ten odd numbers

Evaluate S_{10}.

Answer to Problem 2 (c).

$S_{10} = 10^2 = 100$ (no solution is needed, so only the answer is given).

(d)

How many consecutive odd numbers, starting from 1, must be added for the sum to exceed 1000?

Solution: We need the smallest n with $n^2 > 1000$, that is $n > \sqrt{1000} \approx 31.6$.

Answer to Problem 2 (d).

32 odd numbers.

Verification: The preceding sum is $S_{31} = 961$, while $S_{32} = 1024$. Thus 32 is the first value that exceeds 1000.

Problem 3.

A right triangle has legs $a = 3$ and $b = 4$, as shown in Fig. 1.2. Find the length of the hypotenuse c and the angle θ opposite the leg a.

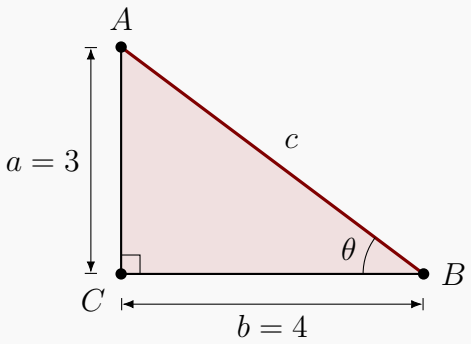


Figure 1.2: Right triangle with legs a and b.

Solution: By the Pythagorean theorem,

$$c = \sqrt{a^2 + b^2} = \sqrt{9 + 16} = 5.$$

The leg a is opposite θ and the leg b is adjacent to it, so

$$\tan\theta = \frac{a}{b} = \frac{3}{4}, \qquad \theta = \arctan\frac{3}{4} \approx 36.87^\circ.$$

Answer to Problem 3.

$c = 5$ and $\theta = \arctan(3/4) \approx 36.87^\circ$.

Chapter 2.

Figures and References

Problem 1.

Consider the function $f(x) = \frac{1}{4}x^2 + \frac{1}{2}$, shown in Fig. 2.1, and the point $P = (3, f(3))$ on its graph.

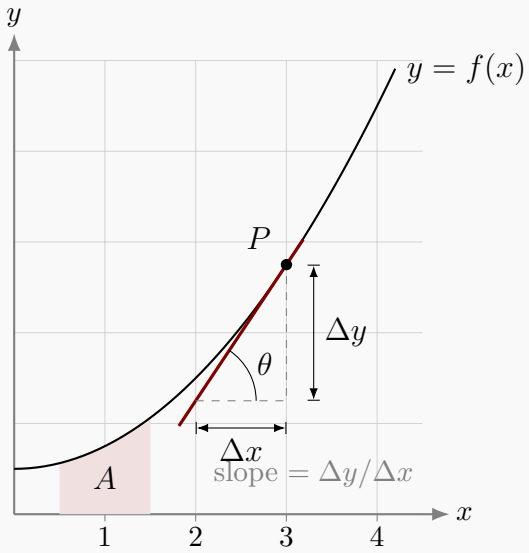


Figure 2.1: Graph of f, its tangent at P and the area A.

(a)

Find the slope of the tangent at P and the angle θ that it makes with the x axis.

Solution: The derivative is

$$\frac{\mathrm{d}f}{\mathrm{d}x} = \frac{x}{2},$$

so the slope at $x = 3$ is $f'(3) = 3/2$. In the slope triangle of Fig. 2.1, $\Delta y/\Delta x = \tan\theta$, hence $\theta = \arctan(3/2)$.

Answer to Problem 1 (a).

Slope $3/2$, and $\theta = \arctan(3/2) \approx 56.31°$.

(b)

Find the equation of the tangent at P.

Solution: Since $f(3) = \frac{9}{4} + \frac{1}{2} = \frac{11}{4}$, the line through P with slope $3/2$ is

$$y = \frac{11}{4} + \frac{3}{2}(x - 3) = \frac{3}{2}x - \frac{7}{4}.$$

Answer to Problem 1 (b).

$y = \frac{3}{2}x - \frac{7}{4}$.

(c)

Find the area A under the graph of f between $x = \frac{1}{2}$ and $x = \frac{3}{2}$.

Solution:

$$A = \int_{1/2}^{3/2} \left(\frac{x^2}{4} + \frac{1}{2} \right) \mathrm{d}x = \left[\frac{x^3}{12} + \frac{x}{2} \right]_{1/2}^{3/2} = \frac{27-1}{96} + \frac{1}{2} = \frac{13}{48} + \frac{24}{48}.$$

Answer to Problem 1 (c).

$A = \frac{37}{48} \approx 0.771.$

Problem 2.

Use Euclid's algorithm to find the greatest common divisor of 84 and 36.

Solution:

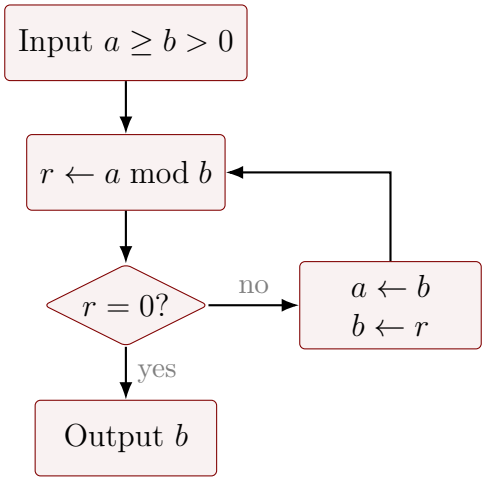


Figure 2.2: Euclid's algorithm.

Starting from $a = 84$ and $b = 36$, each step of the algorithm (Fig. 2.2) replaces (a, b) by $(b, a \bmod b)$:

Step	a	b	$r = a \bmod b$
1	84	36	12
2	36	12	0

The remainder is zero at step 2, so the algorithm outputs $b = 12$. Indeed $84 = 7 \cdot 12$ and $36 = 3 \cdot 12$, with $\gcd(7, 3) = 1$.

Answer to Problem 2.

$\gcd(84, 36) = 12$.

Problem 3.

For which n does the sum S_n of Problem 1.2 satisfy $S_n = 5n - 6$?

Solution: By part 1.2(a) (proved in Chap. 1 Prob. 2), $S_n = n^2$, so the condition becomes $n^2 - 5n + 6 = 0$. This is the equation of Chap. 1 Prob. 1, whose factorization (1.1) gives $n = 2$ or $n = 3$. Both are positive integers, and indeed $S_2 = 4 = 5 \cdot 2 - 6$ and $S_3 = 9 = 5 \cdot 3 - 6$.

Answer to Problem 3.

$n = 2$ or $n = 3$.

Problem 4. *Not done*

Show that the sum of the first n even numbers, $2+4+\cdots+2n$, is $n(n+1)$. (Hint: compare it with S_n, or use induction as in Problem 1.2.)

Problem 5.

Use the two panels in Fig. 2.3 to answer the following questions.

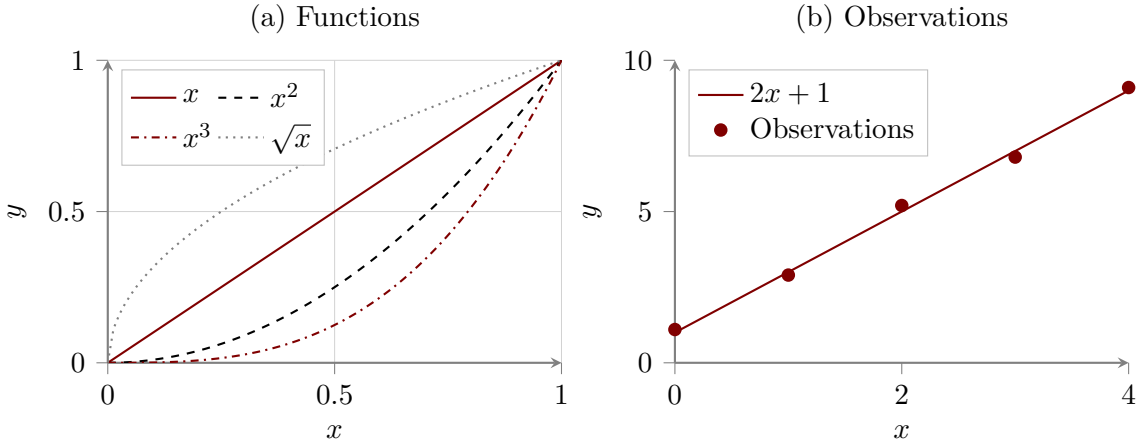


Figure 2.3: Functions on the unit interval and observations of $y = 2x + 1$.

(a)

Order x, x^2, x^3 and $\sqrt{x}$ for $0 < x < 1$.

Solution: Multiplication by $0 < x < 1$ decreases a positive number, so $x^3 < x^2 < x$. Also $x^2 < x$ implies $x < \sqrt{x}$.

Answer to Problem 5 (a).

$x^3 < x^2 < x < \sqrt{x}$.

(b)

What is the largest absolute deviation of the five observations from $y = 2x + 1$?

Solution: The absolute deviations $|y_{\text{measured}} - (2x + 1)|$ at $x = 0, 1, 2, 3, 4$ are 0.1, 0.1, 0.2, 0.2 and 0.1, respectively.

Answer to Problem 5 (b).

The largest absolute deviation is 0.2, at $x = 2$ and $x = 3$.

Chapter 3.

Sections and Progress

3.1 Sequences

Problem 1.

Compute the sum $1 + 2 + 3 + \cdots + 100$.

Solution: Pair the first term with the last, the second with the second to last, and so on: each of the 50 pairs $k + (101 - k)$ adds up to 101.

Answer to Problem 1.

$1 + 2 + \cdots + 100 = 50 \cdot 101 = 5050$.

Problem 2.

An arithmetic sequence starts with $3, 7, 11, \ldots$. Find its tenth term a_{10} and the sum of its first ten terms.

Solution: The common difference is 4, so $a_n = 3 + 4(n-1)$ and $a_{10} = 39$. The sum of an arithmetic sequence is the number of terms times the mean of the first and last terms:

$$a_1 + \cdots + a_n = \frac{n(a_1 + a_n)}{2}. \tag{3.1}$$

Answer to Problem 2.

$a_{10} = 39$, and the sum is $\frac{10(3+39)}{2} = 210$.

Problem 3. *Not done*

Find a closed form for the sum of the first n squares, $1^2 + 2^2 + \cdots + n^2$.

Problem 4. *Not done*

Show that the sequence $a_n = \left(1 + \frac{1}{n}\right)^n$ is increasing and bounded above by 3.

Solution: By the binomial theorem,

$$a_n = \sum_{k=0}^{n} \binom{n}{k} \frac{1}{n^k} = \sum_{k=0}^{n} \frac{1}{k!} \prod_{j=0}^{k-1} \left(1 - \frac{j}{n}\right).$$

Each product grows with n, and a_{n+1} has one more positive term than a_n, so the sequence is increasing. The bound remains to be proved.

§12 Series

Problem 1.

Find the sum of the geometric series $1 + \frac{1}{2} + \frac{1}{4} + \dots$.

Solution: The partial sums are

$$S_n = \sum_{k=0}^{n-1} \frac{1}{2^k} = \frac{1 - 2^{-n}}{1 - \frac{1}{2}} = 2\left(1 - 2^{-n}\right),$$

and $2^{-n} \to 0$.

Answer to Problem 1.

The sum is 2.

Problem 2. *Not done*

Consider the series $\sum_{k=1}^{\infty} \frac{1}{k(k+1)}$.

(a)

Write $\frac{1}{k(k+1)}$ as a difference of two simple fractions.

Solution: Since $(k+1)-k=1$,

$$\frac{1}{k}-\frac{1}{k+1}=\frac{(k+1)-k}{k(k+1)}=\frac{1}{k(k+1)}.$$

Answer to Problem 2 (a).

$\frac{1}{k(k+1)}=\frac{1}{k}-\frac{1}{k+1}$.

(b) *Not done*

Compute the partial sum $S_n=\sum_{k=1}^{n}\frac{1}{k(k+1)}$.

(c)

Deduce the sum of the series.

Answer to Problem 2 (c).

The sum is 1.

Problem 3.

Show that the series $\sum_{k=1}^{\infty} \frac{1}{k^2}$ converges.

Solution: For $k \geq 2$, $\frac{1}{k^2} < \frac{1}{(k-1)k}$. By part §12.2(a) (§12 Prob. 2), these terms telescope:

$$\sum_{k=1}^{n} \frac{1}{k^2} < 1 + \sum_{k=2}^{n} \left(\frac{1}{k-1} - \frac{1}{k} \right) = 2 - \frac{1}{n} < 2.$$

The partial sums increase and are bounded, so the series converges.

Answer to Problem 3.

The series converges, and its sum is less than 2.

Problem 4.

Find the sum of the first 20 terms of the sequence of Problem 3.1.2, and compare it with the sum of the first 20 odd numbers.

Solution: By (3.1) (Sec. 3.1 Prob. 2), with $a_{20} = 3 + 4 \cdot 19 = 79$, the sum is $\frac{20(3+79)}{2} = 820$. The sum of the first 20 odd numbers is $20^2 = 400$ (Chap. 1 Prob. 2), and the series of Problem §12.1 shows that sums of infinitely many terms can still be finite.

Answer to Problem 4.

820, against 400 for the odd numbers.

Chapter 4.

Showcase

Problem 1. Sectors and areas *Not done*

Fig. 4.1 shows the point P at the angle θ and the area A.

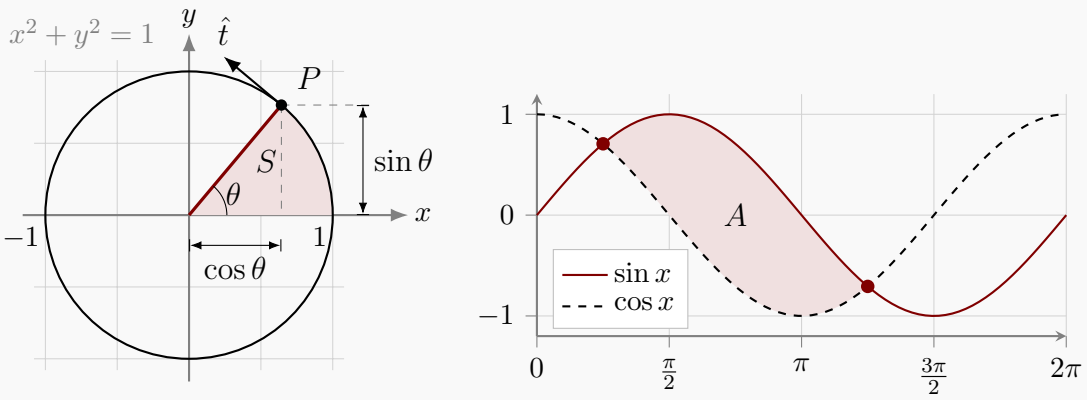


Figure 4.1: Left: the sector S of angle θ. Right: the area A between $\sin x$ and $\cos x$.

(a) Area of a sector

Find the area of the sector S.

Solution: The sector is the fraction $\theta/2\pi$ of the disk, of area πr^2:

$$S = \frac{\theta}{2\pi}\,\pi r^2 = \frac{\theta r^2}{2}. \tag{4.1}$$

Answer to Problem 1 (a).

$S = \theta/2$ for $r = 1$.

(b)

Find the area A.

Solution: The curves cross at $x = \pi/4$ and $5\pi/4$. As in Chap. 2 Prob. 1,

$$A = \int_{\pi/4}^{5\pi/4} (\sin x - \cos x)\,dx = \Big[-\cos x - \sin x\Big]_{\pi/4}^{5\pi/4} = \sqrt{2} + \sqrt{2}.$$

Answer to Problem 1 (b).

$A = 2\sqrt{2} \approx 2.83$.

Verification: As $\sin x - \cos x = \sqrt{2}\sin(x - \pi/4)$, A is $\sqrt{2}$ times one arch of sin, of area 2: the sector (4.1) of part 4.1(a) with $\theta = 4\sqrt{2}$.

(c) *Not done*

Show that $\sin^2\theta + \cos^2\theta = 1$ from the lengths in Fig. 4.1.